

Latency-Aware Unbeatable RPS

Early Hand-Motion Prediction
from Real-Time Hand Landmarks



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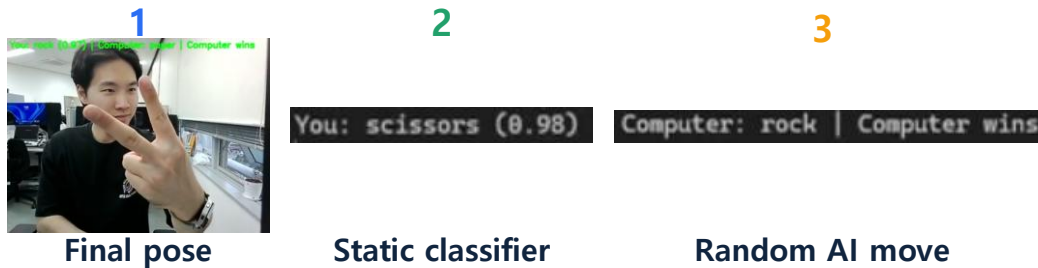
Motivation: Static Recognition Is Too Late

The project reframes RPS from image classification into a timing-sensitive perception-to-action problem.

BASELINE

Recognize a completed pose

Single frame → static class → random AI move

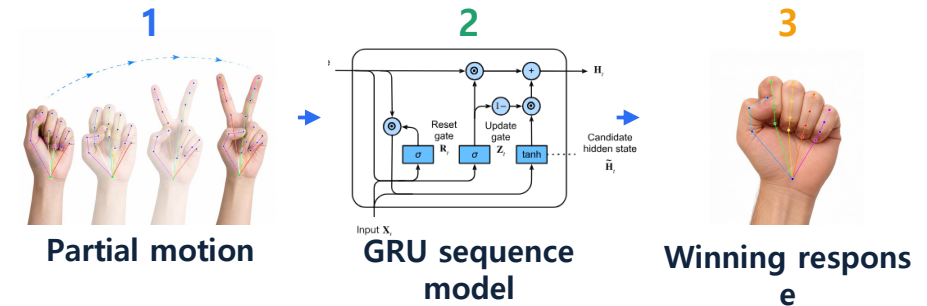


Good for recognition, weak for winning.

PROPOSED

Predict what the motion is becoming

Live landmarks → temporal prediction → counter-move

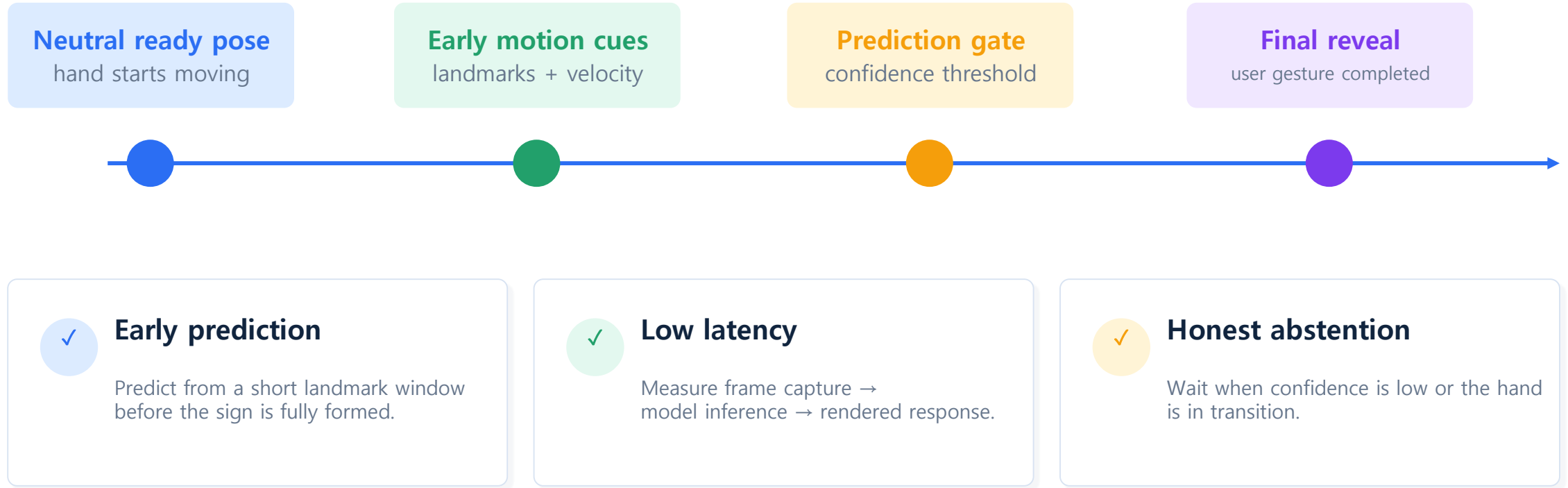


Winning requires timing, not just accuracy.

Key shift: from “What hand sign is visible now?” to “What hand sign is this motion becoming?”

Problem Statement: Win Under Latency Constraints

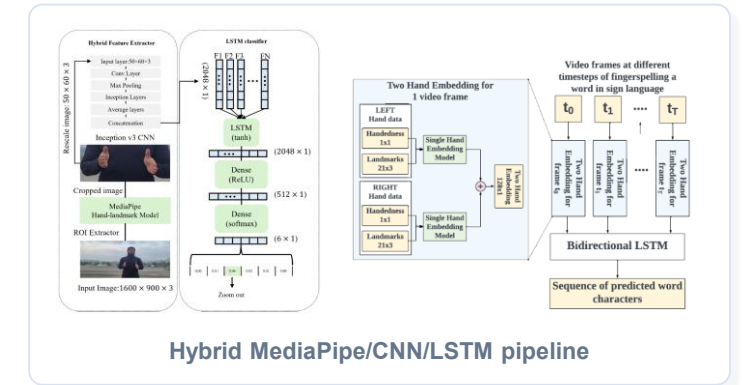
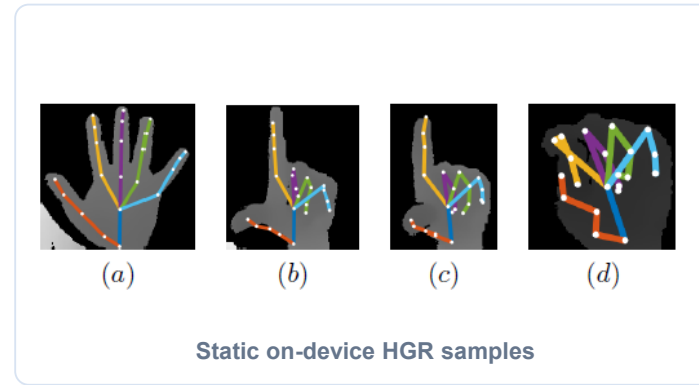
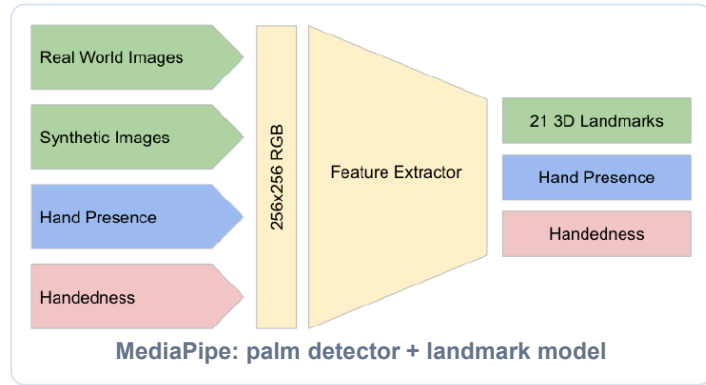
The system should only act when it can predict confidently and respond fast enough.



Unbeatable is the demo concept — measured win rate is the claim to validate.

Related Work: What This Project Borrows

Existing work supports real-time landmarks, on-device gesture recognition, and temporal modeling.



① Real-time hand landmarks

Use MediaPipe Hand Landmarker to avoid training a detector from scratch.

② Static gesture baseline

Closed_Fist, Open_Palm, and Victory map naturally to RPS.

③ Custom gesture adaptation

Small custom datasets can support task-specific gesture heads.

④ Temporal prediction

LSTM/GRU-style sequence models capture motion before final pose.

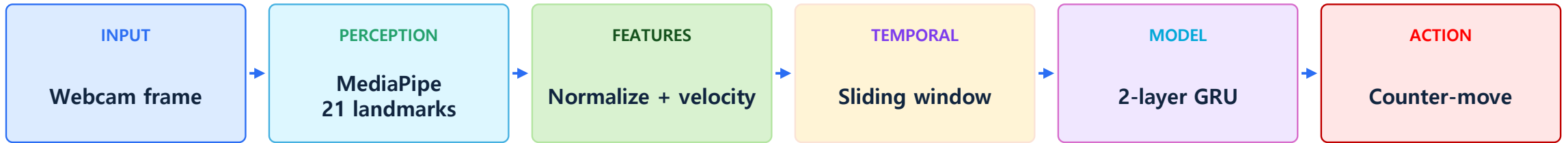
Zhang, Fan, et al. "Mediapipe hands: On-device real-time hand tracking." *arXiv preprint arXiv:2006.10214* (2020).

Sung, George, et al. "On-device real-time hand gesture recognition." *arXiv preprint arXiv:2111.00038* (2021).

Yaseen, et al. "Next-gen dynamic hand gesture recognition: Mediapipe, inception-v3 and lstm-based enhanced deep learning model." *Electronics* 13.16 (2024): 3233.

Proposed System Overview

A live webcam loop turns partial hand motion into an immediate winning counter-move.



Feature vector

- wrist-centered x,y,z
- palm-size normalization
- first-order velocity
- optional handedness flag

Goal: robust to hand position and camera distance

Prediction classes

rock

paper

scissors

neutral

Neutral class prevents forced wrong moves during ambiguous motion.

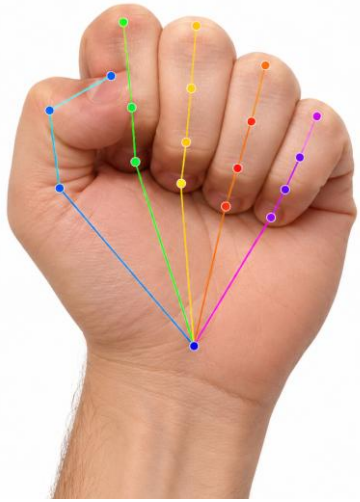
Live overlay

- hand landmarks
- predicted gesture
- confidence score
- decision latency
- AI counter-move

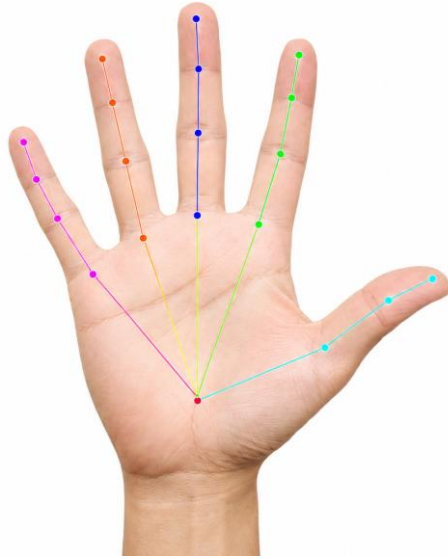
All timestamps are logged for evaluation.

More Diverse Hand-Tracking Examples

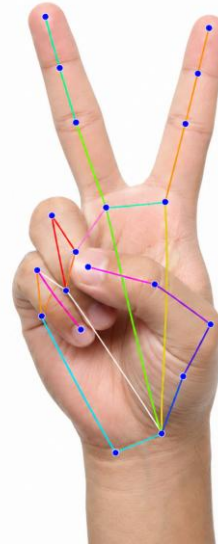
Use multiple landmark examples so the proposal does not depend on one repeated tracking image.



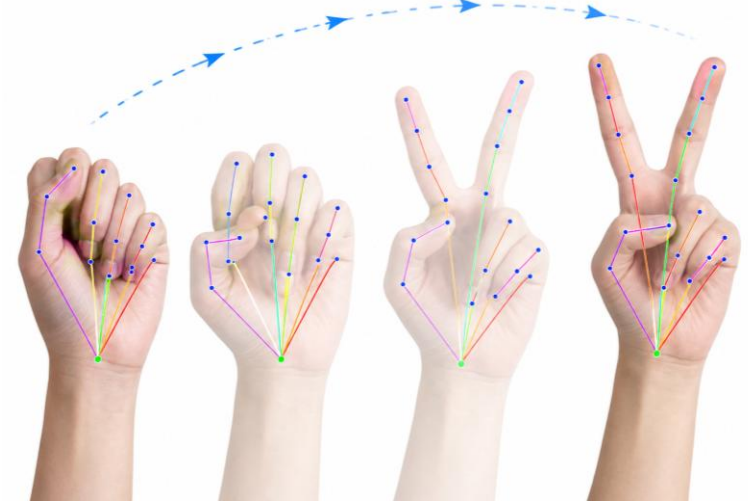
Rock / Closed Fist



Paper / Open Palm



Scissors / Victory



Transition sequence

These examples cover closed hand, open hand, two-finger hand shape, and multi-frame transition the exact diversity needed for early motion prediction.

Timeline and Team Responsibilities

The proposal prioritizes an end-to-end demo first, then improves accuracy and latency.

Before Progress Update 1

Literature review · MediaPipe extraction · Teachable Machine baseline

Before Progress Update 2

First dataset · Initial GRU model · Static-vs-temporal comparison · Overlay prototype

Before Final Submission

Threshold tuning · latency/win-rate evaluation · demo video · final report/slides

Timeline



Proposal



Prototype



Final demo

Cho Min Kyu

model design · real-time inference ·
latency/win-rate evaluation · demo integration

Kim Seung Hwan

data collection protocol · video/landmark
dataset · model training experiments

Expected Deliverables and Conclusion

The final output is a measurable real-time demo, not a vague “AI wins” claim.

Real-time webcam app

landmarks, confidence, latency, AI counter-move

Trained sequence model

landmark-window GRU classifier

Evaluation logs

accuracy, early prediction, latency, win rate, abstention

Demo video + report

comparison with static baselines and failure analysis

Core contribution

Turn a simple webcam RPS game into a compact embodied AI timing problem: perception, temporal prediction, confidence gating, and latency-aware action selection.



Thank You
VOICE Lab